Legacy Carry - Firebase Push Notification & PDF Guide

FCM HTTP v1 & Laravel 11

Amenity Booking, Polls & Voting, Resident Reports, and Laravel PDF Export

1. Firebase FCM HTTP v1 Standard Body Structure

Firebase ke latest HTTP v1 API me notification aur additional payload is format me bheja jata hai:

```
{
  "message": {
    "token": "RECIPIENT_DEVICE_FCM_TOKEN",
    "notification": {
      "title": "Notification Title",
      "body": "Notification message summary for resident/admin",
      "image": "https://your-domain.com/storage/banners/photo.jpg"
    },
    "data": {
      "type": "event_type_key",
      "id": "123",
      "click_action": "FLUTTER_NOTIFICATION_CLICK"
    },
    "android": {
      "priority": "high",
      "notification": {
        "channel_id": "legacy_carry_channel",
        "sound": "default"
      }
    },
    "apns": {
      "payload": {
        "aps": {
          "sound": "default",
          "mutable-content": 1
        }
      }
    }
  }
}
```

2. The 3 Core Notification Payloads

📌 Case 1: Amenity Booking (Confirmed / Pending / Rejected)

Jab resident amenity book karta hai ya admin approve/reject karta hai:

```
{
  "message": {
    "token": "USER_FCM_TOKEN",
    "notification": {
      "title": "Amenity Booking Confirmed! 🎉",
      "body": "Your booking for Swimming Pool on 2026-10-25 (04:00 PM - 05:00 PM) is confi
    },
    "data": {
      "type": "amenity_booking_confirmed",
      "booking_id": "42",
      "amenity_id": "5",
      "status": "confirmed",
      "screen": "amenity_booking_details",
      "click_action": "FLUTTER_NOTIFICATION_CLICK"
    }
  }
}
```

🗳️ Case 2: Society Poll / Vote (New Poll / Vote Cast / Results)

Jab naya poll publish hota hai ya poll close hone par results aate hain:

```
{
  "message": {
    "token": "USER_FCM_TOKEN",
    "notification": {
      "title": "New Society Poll: Diwali Celebration Budget 🗳️",
      "body": "A new poll is live! Cast your vote before 28th October."
    },
    "data": {
      "type": "new_poll_published",
      "poll_id": "18",
      "screen": "poll_details",
      "click_action": "FLUTTER_NOTIFICATION_CLICK"
    }
  }
}
```

📝 Case 3: Resident Report / Complaint (Created & Status Updates)

Jab resident koi issue/complaint submit karta hai ya admin uska status update karta hai:

```
{
  "message": {
    "token": "RESIDENT_FCM_TOKEN",
    "notification": {
      "title": "Report Status Updated: Resolved ✅",
      "body": "Your report #RPT-20261021-AB12 (Water Leakage) has been resolved."
    },
    "data": {
      "type": "resident_report_status",
      "report_id": "89",
      "reference": "RPT-20261021-AB12",
      "status": "resolved",
      "screen": "report_details",
      "click_action": "FLUTTER_NOTIFICATION_CLICK"
    }
  }
}
```

3. Laravel Backend PHP Integration

Aapke codebase me `SocietyNotificationService` aur `FCMHelper` already integrated hain:

```
// 1. Amenity Booking Notification
\App\Services\SocietyNotificationService::notifyUser(
    $residentUser,
    'Amenity Booking Confirmed',
    "Your booking for {$amenity->name} on {$date} is confirmed.",
    'amenity_booking_confirmed',
    ['booking_id' => (string) $booking->id, 'amenity_id' => (string) $amenity->id]
);

// 2. Poll Notification to All Society Residents
\App\Services\SocietyNotificationService::notifySocietyResidents(
    $societyId,
    'New Society Poll: ' . $poll->title,
    'A new poll has been opened for voting. Cast your vote now!',
    'new_poll_published',
    ['poll_id' => (string) $poll->id]
);

// 3. Resident Report Update Notification
\App\Services\SocietyNotificationService::notifyUser(
    $residentUser,
    'Report Status: ' . ucfirst($report->status),
    "Your report {$report->reference} has been marked as {$report->status}.",
    'resident_report_status',
    ['report_id' => (string) $report->id, 'reference' => $report->reference, 'status' => $repo
);
```

4. PDF Generation in Laravel (Resident Report & Amenity Booking)

Reports ya Amenity receipts ka PDF export karne ke liye sabse recommended package `barryvdh/laravel-dompdf` hai:

Step 1: Install Package

```
composer require barryvdh/laravel-dompdf
```

Step 2: Controller PDF Export Method

```
use Barryvdh\DomPDF\Facade\Pdf;
use App\Models\Report;

public function downloadReportPdf($id)
{
    $report = Report::with('user')->findOrFail($id);

    // Blade view se PDF render karein
    $pdf = Pdf::loadView('admin.reports.pdf_template', compact('report'));

    return $pdf->download("Report-{$report->reference}.pdf");
}
```

Step 3: Route Setup

```
Route::get('/admin/reports/{id}/pdf', [ReportController::class, 'downloadReportPdf'])
    ->name('admin.reports.pdf');
```